INCH-POUND

CCC-C-476H
January 29, 1990
SUPERSEDING
CCC-C-476G
October 21, 1983

FEDERAL SPECIFICATION

CLOTH, BUNTING, NYLON; AND NYLON AND WOOL

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nylon and nylon-and-wool cloth (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The finished cloth shall be of the following types and classes, as specified (see 6.2)

Type I - 100 percent nylon filament

Class A - Lightweight
Class B - Heavyweight

Type II - 75 percent nylon (staple) and 25 percent wool

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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Federal Specification

PPP-P-1132 - Packaging of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics
PPP-B-1133 - Packaging of Synthetic Fiber Fabrics

Federal Standards

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA, Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Federal Regulations:

U.S. Department of Agriculture

Methods of Test for Grades of Wool

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402.)

U.S. Government Printing Office

Rules and Regulations Under the Wool Products Labeling Act

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402-0001. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

2 2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

American Association of Textile Chemists and Colorists (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Nylon (type I only) For type I bunting the material shall be bright, high tenacity, and continuous nylon filament. Testing shall be as specified in 4.4.3.

3.3.2 Nylon (type II only). For type II bunting the material shall be bright staple nylon The material shall not contain waste, reused, nor reclaimed nylon fiber. Testing shall be as specified in 4.4.3.

3.3.3 Wool (type II only). The wool top for type II bunting shall be fleece wool, pulled wool, or any combination thereof not lower in grade than 44s U.S. Standard (see 2.1). The use of noils, laps, or any other manufacturing by-product is prohibited (see 4.4.1.1). Testing shall be as specified in 4.4.3.

3.4 Yarn.

3.4.1 Type I bunting.

3.4.1 1 Class A. The warp yarn for class A cloth shall be single end 70 nominal denier. The filling yarn shall be two-ply 70 nominal denier or 140 nominal denier singles, when tested as specified in 4.4.3.

3.4.1.2 Class B. The yarn for class B cloth shall be either 200 or 210 nominal denier single yarns in both warp and filling when tested as specified in 4.4.3.

3.4.2 Type II bunting. Yarns for type II bunting shall be a blend of 75 percent nylon staple and 25 percent wool when tested as specified in 4.4.3 A variation of plus or minus 5 percent content for either fiber will be permitted. The yarns shall be spun from combed top on either the Bradford, French, or American System. The yarn shall be two-ply for both warp and filling, when tested as specified in 4.4.3.

3.5 Physical requirements. The bunting shall meet to the physical requirements specified in table I when tested as specified in 4.4.3.

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3.5.1 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum acceptable width, inclusive of selvages when woven on the fly shuttle looms and exclusive of selvages when woven on shuttleless looms. The selvage for type I cloth shall be not more than 1/4 inch wide and for type II cloth the selvage shall be 1/4 ± 1/8 inch wide

TABLE I. Physical requirements

Type	Class	Weight per sq. yard (minimum) ounces	Yarns per inch (minimum)		Breaking strength pounds (minimum)	
			Warp	Filling	Warp	Filling
I	A	2.5	106	76	125	155
	B	3.6	72	56	225	152
II	-	4.8	32	30	115	100

3.5.2 Weave. The body of the cloth shall be plain weave. The selvages shall be an oxford weave with two ends weaving as one. When shuttleless looms are used, the requirement for selvage shall not apply.

3.6 Color. The color shall be as specified (see 6.2), and either by cable numbers or names, they shall match the applicable color card (see 2.2) or standard sample

3.6.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.6.2 Colorfastness. The finished cloth shall show fastness to light, fresh water, and salt water equal to or better than the standard sample or equal to or better than a rating of "good." The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not lower than 3.5. Testing shall be as specified in 4.4.3.

3.7 Finish.

3.7.1 Type I. The finished cloth shall have a hand equal to the standard sample.

3.7.2 Type II. The cloth shall be crabbed and sheared on both sides and have a hand equal to or better than the standard sample.

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3.8 pH. The pH value of the water extract for type I cloth shall be no less than 5.0 nor more than 8.5 and the value for type II shall be no less than 4.0 nor more than 8.0 when tested as specified in 4.4.3.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths of not less than 40 yards each. When procurement is made by the Government, each length shall be put-up on full width rolls as specified in PPP-P-1132 for type II cloth and PPP-P-1133 for type I cloth.

3.10 Fiber identification. Each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.10.1 Wool content label (type II only). Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act.

3.11 Workmanship. The cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4 1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.4 and tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Wool. The wool shall be examined for grade (see 3.3.3) in undyed top form by comparison with the applicable U.S. Standard (see 2.1). In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be 1 yard of the undyed or untreated top prior to dyeing and blending with nylon fibers The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 pound. The result from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade.

Lot size (pounds)	Sample size (sample unit)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.1.2 Component and material certification. A certificate of compliance may be acceptable as evidence that the cloth conforms to the requirements specified in 3.3.1, 3.3.2, 3.3.3, 3.4.1.1, 3.4.1.2, and 3.4.2.

4.4.2 End item visual examination.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on one side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (sample unit) 1/
3200 or less	10
3201 up to and including 10,000	15
10,001 and over	25

1/ Not more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

All defects as defined in Sections II and III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1. Only slubs and knots of type II cloth which exceed the limits shown in figure 1 of FED-STD-4 shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following point values:

35 points for type I cloth
30 points for type II cloth

The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceeds the following point values:

53 points for type I cloth
45 points for type II cloth

If one roll exceeds the point level per 100 square yards, a second sample shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2 1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

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The following defects, when present, shall be scored four points for each yard in which they occur.

Shading defects - side to side, side to center, or end to end
Poor dye penetration - mottled cloudy, or streaky
Moire condition, intermittently, or continuously throughout piece
Not evenly and closely sheared on both sides (type II)
Overall uncleanness
Width beyond specified tolerances

4.4.2 2 Length Examination.

4 4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be more than 2 yards less than the length marked on the roll ticket, or any roll containing more than one piece, or any roll containing less than 40 yards or as required shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length

4.4.2.2.2 Examination for total yardage. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample (see 4 4.2.2.1) is less than the total of the length marked on the roll tickets.

4.4 2.3 Roll marking examination During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the requirements. The lot shall be unacceptable if two or more rolls in the sample are not labeled, ticketed in accordance with the Wool Labeling Act and the Textile Fiber Products Identification Act.

4.4.2.4 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll in the sample off shade or shaded side to side, side to center, or end to end, and any roll that does not have the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.4.3 End item testing The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 1-1/2 yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The lot size shall be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final result. The smaple size shall be as follows:

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Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Nylon identification	3.3.1, 3.3.2	1530 1/
Wool identification (type II)	3.3.3	1100 1/
Luster	3.3.1, 3.3.2	Visual 1/
Tenacity	3.3.1	1/
Absence of nylon waste, reused or reclaimed fiber	3.3.2	Visual 1/
Denier (type I)	3.4.1.1, 3.4.1.2	1/
Fiber blend content (type II):		
Nylon	3.4.2	2530
Wool	3.4.2	2530 2/
Type of combed top	3.4.2	1/
Yarn ply:		
Warp	3.4.1.1, 3.4.1.2, 3.4.2	Visual 1/
Filling	3.4.1.1, 3.4.1.2, 3.4.2	Visual 1/
Weight	3.5	5041
Yarns per inch	3.5	5050
Breaking strength	3.5	5100
Weave	3.5.2	Visual 3/
Colorfastness to:		
Light	3.6.2	5660
Crocking	3.6.2	5651
Water	3.6.2	5630
Salt water	3.6.2	5632
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The wool shall be calculated as follows:

$$\text{Wool content percent} = \frac{R}{S} \times 100$$

R = Weight of residual fibers
S = Weight of original specimen

3/ One determination shall be made from each sample unit and the results reported as "pass" or "fail."

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4.4.4 Packaging examination. The examination shall be made in accordance with the quality assurance provisions of PPP-P-1133 for type I cloth, and PPP-P-1132 for type II cloth, as applicable.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial preservation. The cloth shall be put-up on rolls and preserved in accordance with the applicable requirements of PPP-P-1132 for type II cloth and PPP-P-1133 for type I cloth.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2)

5.2.1 Levels A, B, and Commercial packing. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132 for type II cloth and PPP-P-1133 for type I cloth.

5.3 Marking. In addition to any special marking specified in the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1132 for type II cloth and PPP-P-1133 for type I cloth.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nylon cloth and nylon and wool cloth are primarily used in the manufacture of various types of flags.

6.2 Acquisition requirements. Acquisition documents must specify the following.

a. Title, number, and date of this specification.
b. Type and class required (see 1.2.1).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Width required (see 3.5.1).
e. Color required (see 3.6).
f. Minimum length if other than specified (see 3.9)
g. Levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Flags
Textile fabrics

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

MILITARY INTERESTS:

Custodians.

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - IH
Navy - MC
Air Force - 82
DLA - CT

User activities.

Navy - SH, OS
Air Force - 45

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HHS - NIH
VA - OSS
JUS - FPI

PREPARING ACTIVITY:

Army - GL

(Project 8305-0312)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
CCC-C-476H	CLOTH, BUNTING, NYLON; AND NYLON AND WOOL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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